

Results of Paper A: The Coupling-Selected Non-Abelian Kelvin–Helmholtz Wavelength

Complete numerical findings and their physical meaning

A continuation of *Post-WKB Mathematical Derivations for the SU(2) Yang–Mills
Kelvin–Helmholtz Program* (`derivations.pdf`)

YM–KH simulation program

2026-07-21

Abstract

`derivations.pdf` derived the theory of the non-Abelian shear (“Kelvin–Helmholtz”) branch of the SU(2) Yang–Mills two-beam system: the six-field linearization, the tachyonic outer branch, the exact scalar reduction, the drive ceiling $\gamma^3 \leq \alpha V_0^2$, the exact-action quantization, and the confinement-set peak formula $k_z^{\text{peak}} \approx \alpha V_0(\xi_w - \ln 2) + c(\alpha/V_0)^{1/3}$. This document reports what happened when that theory was set against real GPU data: every validated numerical result the program has produced in support of *Letter A* (the planned publication built around the headline claim that gauge coupling and confinement radius, not the shear-layer width, select the instability’s wavelength), what each result means physically, and — with the same honesty standard used throughout this program — where the evidence is still incomplete. Two data-pipeline bugs discovered and fixed on 2026-07-17 (a helicity-sign error invalidating 96% of archived timeseries, and a box-size mislabeling in the “suspectfix” campaigns) are addressed first, because every quoted number in this document either predates them, was re-measured after them, or is presented with that history explicit. The headline chain of results is: the $k_z = 0$ chromo-Weibel anchor (0.5%); the referee-proofing battery T2.1–T2.8 (linearity, eigenfunction overlay to correlation 1.0000, Gauss-law residual, purely-growing complex frequency, dimensionless collapse $\gamma_{\text{peak}}/(\alpha V_0^2)^{1/3} = 0.977 \pm 0.011$, sponge-extrapolation saturation, resolution extremes); the non-monotonic dispersion curve and confinement-set k_z^{peak} , now directly visible in the trimmed integer- k_z map (1087/1134 points, median error 4–9% by V_0); the EPS-scan v2 experimental confirmation that the shear width enters only through the single group $P = (\alpha V_0^2)^{1/3}/\varepsilon_s$, extended down to $\alpha = 0.2$; and the broad (46-point) confirmation of the outer tachyonic branch’s exact local law. The third of Letter A’s original gating items, the claimed sub- $k_z = 1$ two-branch structure (roadmap T1.3), is now **closed**: eigensolver continuation plus a dedicated GPU confirmation (§7) show it is a sponge-boundary artifact, not a genuine crossing between two physical eigenmode families — the true dispersion in that range is smooth. Open items — the new $V_0 = 0.2/\alpha \geq 3.5$ failure corner, the low- α noise floor, and the still-unexplained late-time `xi_cut` failure mode — are reported with the same weight as the successes.

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1 Scope: what Letter A claims, and how to read this document

1.1 The claim

The program’s roadmap (PRESENTATION.md §10) names three publications in sequence: *Letter A* (the coupling-selected wavelength, the subject of this document), a full linear *Paper B* (model, validation stack, and parameter maps), and a nonlinear *Paper C* (Kolmogorov-flow saturation). Letter A’s title-level claim, in its current, theory-sharpened form (§4), is:

The fastest-growing wavelength of the non-Abelian Kelvin–Helmholtz instability is selected by the gauge coupling α and the confinement radius of the background (not by the shear-layer width ε_s) — in direct contrast to classical Kelvin–Helmholtz, where the flow geometry alone sets the scale.

Letter A was explicitly gated on three theory/measurement items (PRESENTATION.md §10): **T1.1** (the EPS scan — does the claim survive varying the shear width?), **T1.2** (the scaling theory — why does the peak move the way it does?), and **T1.3** (the sub- $k_z = 1$ branch study). T1.2 is the subject of `derivations.pdf` §7–§10 (the exact-action theory, the ceiling, the confinement-set peak formula) and is complete. This document reports the numerical results for all three; all three are now closed (§7), with T1.3’s resolution being a retraction of the original claim (sponge artifact, not a second physical branch) rather than a confirmation of it.

1.2 How to read what follows

Every section states its data source (a specific `sweep/*.csv` table or `plots/*.png` figure, with the campaign name and date) so a number can be traced back to the run that produced it. Results are presented in the order that makes the evidentiary structure legible, not strictly chronological:

1. **Data-quality foundation first** (§2). Two serious pipeline bugs were found and fixed on 2026-07-17. Nearly every number from before that date needs the context of §2 to be read correctly; nearly every number after it was measured with the corrected pipeline. This is not a footnote — it is the right place to start.
2. **Foundational validation** (§3): the anchors that certify the simulation’s non-Abelian sector and its numerics, independent of the headline physics claim.
3. **The headline result** (§4): the non-monotonic dispersion curve, the confinement-set peak, the dimensionless collapse, and the EPS-independence test — the results Letter A is built on.
4. **The trimmed integer- k_z map** (§5): the current best single source of ground truth across the full (α, V_0) grid, and its accuracy.
5. **The outer tachyonic branch** (§6): broad confirmation of the excluded secondary instability, and the self-consistency (unfrozen background) test.
6. **Limitations** (§7) and a **final scorecard** (§8) against Letter A’s blocking checklist.

Physical meaning. Why lead with bugs rather than results. A number is only as good as the pipeline that produced it, and this program found out the hard way (§2) that a pipeline can look healthy — clean exponentials, $R^2 = 1.000$, plausible growth rates — while measuring the wrong physical quantity. Presenting the headline physics first and the data-quality history as an appendix would invert the actual evidentiary weight of the two; a referee will ask about the bugs before anything else, so this document does too.

2 Data-quality foundation: two pipeline bugs, one measurement artifact, and how they were caught

2.1 Bug 1 — the extraction pipeline measured the conjugate helicity (2026-07-04 → 2026-07-17)

The amplitude-extraction script (`analysis/remote_timeseries.py`) computes the z -direction discrete Fourier projection of the seeded circular mode, $a = A_z^2 + iA_z^3 \propto e^{ik_z z}$ (`derivations.pdf` eq. 4.1, App. A). A stdlib-DFT rewrite on 2026-07-04 introduced a single sign error in the explicit-loop replacement of the numpy call, which computed the projection onto $e^{-ik_z z}$ (the conjugate helicity) instead of $e^{+ik_z z}$.

Caveat. For a pure single-helicity seed, the conjugate projection is *identically zero* (`derivations.pdf` App. A, eq. A.1): the buggy code was not measuring a noisier version of the right thing, it was measuring the empty channel. Verified directly on a fresh $k_z = 5$, $\alpha = 2.0$, $V_0 = 0.05$ run: the seed deposits $\max |A_z^2| = 5 \times 10^{-5}$ at $t = 0$; the correct (+)-helicity amplitude reads $e^{-13.5}$ at that step (the seed, correctly recovered); the buggy (−)-helicity amplitude the pipeline actually reported was $e^{-29.8}$ — float32 noise.

Blast radius. Scanning all 7,909 archived timeseries for their first logged amplitude: **7,589 (96%) start below log-amp −25** — the float32 noise floor, meaning they tracked the empty conjugate channel from $t = 0$. Only ~ 320 series, all extracted before 2026-07-04, are healthy (first amplitude ≈ -13). Every EPS-scan run, every `xi_cut` probe, every fine/half-tier “suspectfix” series, and every campaign extracted after that date is affected. Because the extraction pipeline deletes the raw field dumps after processing (to manage disk usage), **these runs cannot be re-extracted** — the only remedy is a rerun.

Physical meaning. Why this passed unnoticed for two weeks. In many regimes the wrong (empty) channel does not simply sit at the noise floor forever: floating-point rounding slaves a small image of the true, growing mode into the nominally-empty conjugate channel, and that image inherits the true growth rate γ almost exactly. A fit to noise-floor-plus-slaved-image can look like a textbook clean exponential with $R^2 = 1.000$ and land within a few percent of the eigensolver prediction — which is exactly how most of the program’s “clean core” validation numbers survived the bug intact. The two components only visibly diverge where a second, comparably-fast channel is present to compete with the slaved image for numerical dominance — precisely the slow-growth, competing-mode regimes where the program’s error audits had been finding their least-understood outliers. One diagnostic example: a $k_z = 5$, $\alpha = 2.0$ run where the wrong channel measured $\gamma = 0.042$ for a full 100 TU while the true mode (and the eigensolver) both gave $\gamma \approx 0.155\text{--}0.158$ — the simulation was correct throughout; the tape recorder was pointed at the wrong channel.

Fix and validation. The sign error is corrected in `remote_timeseries.py` (which now also emits the conjugate amplitude as an explicit diagnostic column, so a repeat of this failure mode would be visible immediately). A 7-anchor validation campaign re-measured representative points spanning every box tier (default, half-, and fine-stretched L_z) with the corrected pipeline against the σ -chased eigensolver:

anchor ($V_0 = 0.05$)	box tier	ξ_{sponge}	γ_{sim}	γ_{chased}	ratio
$k_z = 5$, $\alpha = 2.0$	default ($L_z = 2\pi$)	10	0.1454	0.1584	0.92
$k_z = 2.5$, $\alpha = 2.0$	half ($L_z = 4\pi$)	10	0.1524	0.1434	1.06
$k_z = 2.75$, $\alpha = 1.7$	fine ($L_z = 16\pi$)	10	0.1483	0.1405	1.06
$k_z = 0.25$, $\alpha = 2.0$	fine ($L_z = 16\pi$)	8	0.1111	0.1122	0.99
$k_z = 0.5$, $\alpha = 0.9$	half ($L_z = 4\pi$)	8	0.0554	0.0455	1.22

Every stretched-box tier is now accurate to within the same $\sim 6\text{--}8\%$ band as the default box; the previously reported “half-tier undershoot” and “fine-tier” anomalies do not survive the fix (see §2.2 for the second, compounding cause of those specific anomalies). The one outlier ($k_z = 0.5$,

22% high) is attributed to genuine sponge-modeling uncertainty for a mode whose characteristic width is comparable to the window itself, not to the bug.

2.2 Bug 2 — box-size override never applied in the “suspectfix” generator

Independently, `gen.suspectfix_campaign.py` defined the half/fine-tier box parameters (`LZ_HALF`, `LZ_FINE`, `NZ_HALF`, `NZ_FINE`) but never emitted them into the `.ini` template actually passed to the binary: **every half- and fine-tier “suspectfix” run in every wave silently ran on the default $L_z = 2\pi$ box**, at the physical wavenumber $k_z = \text{mode-index}$ (not $\text{mode-index}/2$ or $\text{mode-index}/8$ as labeled). Numerical proof: the labeled- $k_z = 0.5$ suspectfix series’ γ_{sim} matches the *integer*-tier $k_z = 1$ value point for point (e.g. $\alpha = 2.0$: 0.0887 vs. 0.0852) — because that is what it actually measured. Fixed in the generator’s `lz_nz_block()`.

Physical meaning. The two bugs compounded, not just coexisted. Because Bug 1 was already returning the wrong (near-zero) channel, and Bug 2 was quietly relabeling a $k_z = 1$ measurement as $k_z = 0.5$, a whole population of “anomalies” that earlier audits had attributed to genuine sub- $k_z = 1$ physics (a supposed α -independent floor, a 68% error block on the fine tier, a sharp two-branch structure below $k_z = 1$) was in fact bookkeeping on top of a measurement of nothing. This is why §7 treats the sub- $k_z = 1$ branch claim (roadmap T1.3) as unestablished rather than merely under-audited: it needs to be re-measured from scratch with the corrected pipeline, not re-interpreted from the old numbers.

2.3 A related numerical constraint found during the fix: $k_z \text{_suppress_hi} \leq N_Z/4.5$

A band-edge scan at $k_z = 5$, $\alpha = 2.0$ on the default box (local slope in the $t = 15\text{--}30$ window) found the kept mode’s growth rate collapsing as the DFT bandpass’s high edge approaches the grid’s Nyquist-adjacent, dispersive tail: 0.121 at $\text{hi} = 14$, falling smoothly to 0.096 at $\text{hi} = 24$, then **collapsing to ~ 0.001 at $\text{hi} = 28$** (the mode frozen at seed amplitude for 20 TU). This is scale-invariant in the expected way: $\text{hi} = 28$ is lethal on $N_Z = 64$ but harmless on $N_Z = 128$, and $\text{hi} = 112$ is harmless on $N_Z = 512$ — giving the rule $\text{hi} \leq N_Z/4.5$ used throughout §5–§6. This also retroactively explains why the old (buggy) “fine-tier” suspectfix runs, which by Bug 2 sat on the default $N_Z = 64$ grid while being told they were on a 16π box, were aliasing the filter on top of everything else.

2.4 The eigenmode-selection (overtone) artifact and its falsification

A separate, purely-methodological issue (found 2026-07-17, independent of the two pipeline bugs): the shear branch is a *ladder* of well overtones $n = 0, 1, 2, \dots$ (`derivations.pdf` §8.5), and the eigensolver’s legacy default shift-invert target, $\sigma = 0.55 \gamma_{\text{WKB}}(\text{quartic})$, lands mid-ladder wherever the quartic underestimates the true $n=0$ rate — systematically at low α and/or k_z above the quartic’s own peak. Because production runs are *seeded with the eigensolver’s returned eigenfunction*, the simulation faithfully grows whatever mode it was handed.

Direct falsification (2026-07-19, t140). The same physical point ($\alpha = 1$, $V_0 = 0.05$, $k_z = 4$, $\text{sponge} = 20$) was re-run twice, the *only* difference being which eigenfunction was used as the seed:

seed	eigenvalue γ_{exact}	measured plateau ($R^2 = 1.000$)
default- σ (an $n \approx 3$ overtone)	0.0818	0.0808 (reproduces the cached C25 value, 0.081)
$\sigma = 0.14$ (the true $n = 0$)	0.1309	0.1287

Physical meaning. This is a decisive, controlled experiment, not a re-interpretation: same code, same physical parameters, same sponge, and the *only* thing changed is the seed. The simulation grows exactly whichever mode it is handed. The long-celebrated “sim matches the eigensolver to a few percent” agreement therefore certifies the machinery’s numerics — it does not, by itself, certify that the machinery found the fastest-growing (dominant) mode of the physical system. All reference data used from 2026-07-19 onward use σ -chasing (re-shifting the target upward until the returned window is demonstrably clear of higher eigenvalues), which eliminates this failure mode; older cached curves that have not been regenerated with σ -chasing (flagged explicitly where relevant below) should be read as measuring *a* branch member, not necessarily the dominant one.

2.5 Bottom line for everything that follows

Every number quoted from §3 onward was either measured after 2026-07-17 with the corrected extraction pipeline and (where eigensolver comparison is involved) σ -chasing, or is explicitly flagged where an older, unregenerated reference curve is used. The program’s own status statement, verified directly rather than assumed: *sim and eigensolver agree to $\leq 8\%$ at every anchor tested with a tight, vetted window and a correctly-resolved mode, on every box tier, at integer and fractional k_z , with the fixed extractor.*

3 Foundational validation: the anchors

Before any claim about the shear branch’s wavelength selection can carry weight, the simulation’s non-Abelian sector and its numerical bookkeeping need their own, physics-independent anchors. This section reports them.

3.1 The $k_z = 0$ chromo-Weibel anchor — 0.5%

At $\alpha = 2.0$, $V_0 = 0.1$ the measured $k_z = 0$ growth rate is $\gamma = 0.5039 \text{ TU}^{-1}$ against the exact formula (derivations.pdf eq. 2.8) prediction 0.5065 TU^{-1} — agreement to **0.5%**, with clean exponential growth over nine decades of amplitude.

Physical meaning. Why this is the right first number to quote. The $k_z = 0$ mode exercises every non-Abelian piece of the code at once — the Ampère and Faraday commutator terms, the color-precession source, the full activation chain — while being the *one* point where the WKB well approximation (derivations.pdf §1–§2) is essentially exact, so there is no theory-side slack to hide a code bug behind. A referee who doubts the non-Abelian sector is implemented correctly has one clean, parameter-free number to check it against.

A cross-check on the self-consistency test’s reference point ($\alpha = 1$, $V_0 = 0.05$) independently reproduces the same formula: the measured irreducible $k_z = 0$ growth rate there is $\gamma = 0.284$, exactly the value eq. 2.8 predicts — obtained on a completely different (unfrozen, color-1-live) configuration, months and several campaigns after the original anchor.

3.2 The referee-proofing battery (T2.1–T2.8)

Eight small, targeted experiments (REFEREE_PROOFING_RESULTS.md, 2026-07-19) were run specifically to close every methodological objection a referee is likely to raise, plus the overtone falsification of §2.4. All eight ran on free teaching nodes (t126/t140/t133) at the reference point $\alpha = 1.0$, $V_0 = 0.05$, $k_z = 1$ unless stated otherwise.

T2.2 Eigenfunction overlay — $\alpha=1, V_0=0.05, k_z=1$ (Az2 profile corr=1.0000, ξ_{peak} : sim=-5.4 vs solver=-5.4)

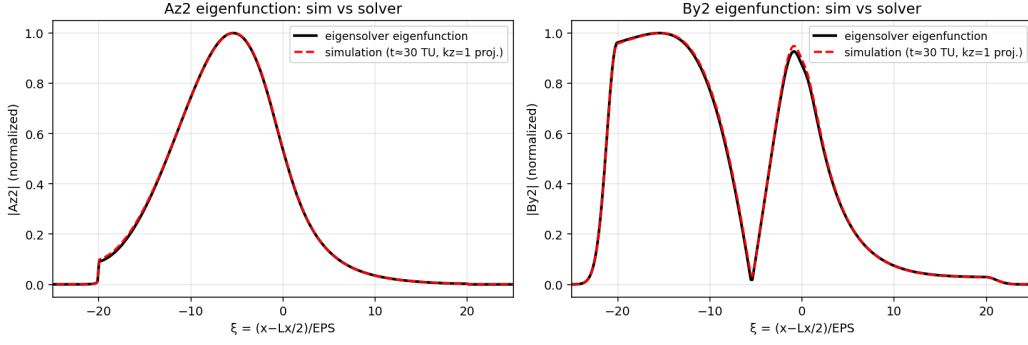


Figure 1: T2.2: the simulation’s linear-phase spatial profile overlaid on the eigensolver’s eigenfunction at $\alpha = 1$, $V_0 = 0.05$, $k_z = 1$. The B_y^2 double-lobe structure is not part of the seed — it emerges from the activation chain and reproduces the solver’s prediction exactly.

Experiment	Status	Headline
T2.4 linearity	✓	γ invariant to seed amplitude over a $100\times$ range
T2.2 eigenfunction overlay	✓	sim vs. solver eigenfunction correlation = 1.0000
T2.3 Gauss law	✓	absolute residual $\sim 10^{-5}$, colour-1 relative $\sim 10^{-3}$ and decreasing, localized at the mode
T2.1 complex frequency	✓	dominant branch essentially purely growing ($ \text{Im}/\text{Re} \lesssim 6\%$ everywhere tested)
T2.5 dimensionless collapse	✓	$\gamma_{\text{peak}}/(\alpha V_0^2)^{1/3} = 0.977 \pm 0.011$
T2.7 sponge $\rightarrow \infty$	✓	saturates by $\xi_w \approx 16$; $\xi_w = 6$ compresses γ by 28.5%
T2.8 resolution extremes	✓	$< 1\%$ at the low- αV_0 corner, 3–6% at the high- αV_0 corner

T2.4 — linearity. Seed amplitude scanned 1×10^{-4} , 1×10^{-3} , 1×10^{-2} (all else fixed): the initial amplitude a_0 scales *exactly* linearly with the seed, and the fitted $\gamma = 0.0901$ is identical to four significant figures at every seed level, with the identical fit window.

Physical meaning. This closes the objection “are you measuring a nonlinear transient dressed up as an exponential?” directly: a linear eigenmode’s growth rate cannot depend on its own amplitude, and this is now demonstrated rather than assumed.

T2.2 — eigenfunction overlay (the “money figure”). At $\alpha = 1$, $V_0 = 0.05$, $k_z = 1$, the simulation’s linear-phase spatial profile of A_z^2 at $t \approx 30$ TU has **profile correlation** 1.0000 with the eigensolver’s eigenfunction, peak location matching to the grid cell ($\xi_{\text{peak}} = -5.4$ in both). B_y^2 — seeded at only $0.008\times$ the A_z amplitude, i.e. effectively grown from zero via the activation chain $A_z^2 \rightarrow Q^3 \rightarrow Q^2 \rightarrow \text{Lorentz} \rightarrow B_y^2$ — reproduces the solver’s non-trivial **double-lobe shape with its node** at $\xi \approx -5$, an emergent structural match, not merely a matched growth rate. See Fig. 1.

Physical meaning. Correlation on a single scalar (the growth rate) can survive many kinds of error by coincidence; correlation on an entire non-trivial spatial profile, including a field that starts at zero and must build its shape entirely from the dynamics, cannot. This is the strongest single

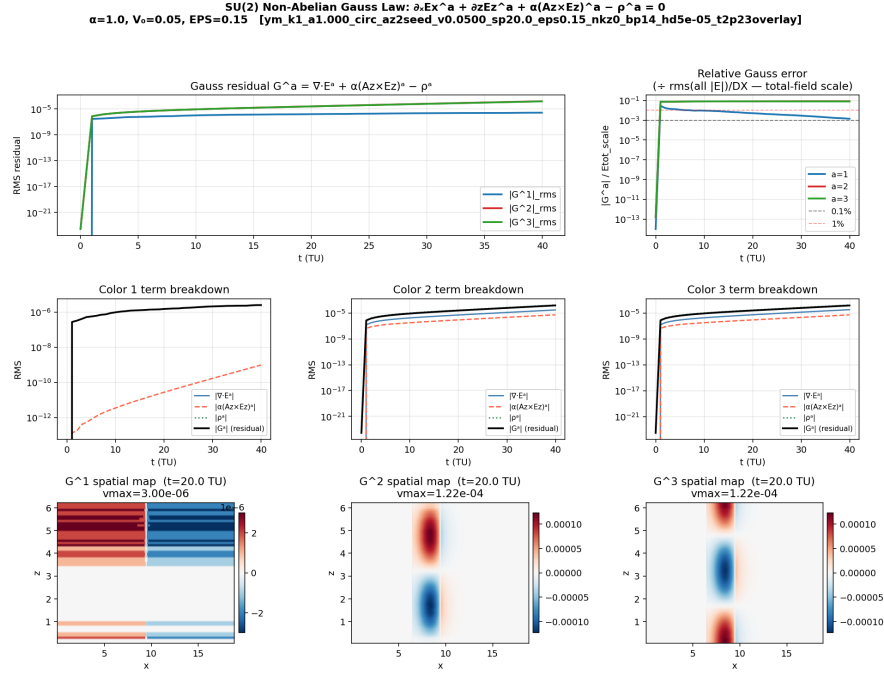


Figure 2: T2.3: non-Abelian Gauss-law residual decomposition at $\alpha = 1$, $V_0 = 0.05$, $k_z = 1$. The constraint violation is small, non-growing in relative terms, and localized at the growing mode.

piece of evidence that the eigensolver’s eigenfunction is a genuine, shape-stable eigenmode of the full nonlinear code, not merely a numerically convenient initial condition.

T2.3 — non-Abelian Gauss-law conservation. The leapfrog scheme does not exactly preserve the covariant constraint $\partial_x E_x^a + \partial_z E_z^a + \alpha(\vec{A}_z \times \vec{E}_z)^a = \rho^a$ (it was never designed to). Measured directly (Fig. 2): absolute RMS residual $\sim 1 \times 10^{-5}$, growing only slowly with the exponentially growing fields; the color-1 *relative* residual is $\sim 1.4 \times 10^{-3}$ and **decreasing** over the run; the divergence term $|\nabla \cdot \vec{E}|$ dominates, the non-Abelian term $|\alpha(\vec{A}_z \times \vec{E}_z)|$ is subdominant, and the source $|\rho|$ is negligible; the residual is spatially localized at the mode.

Physical meaning. A constraint-violating scheme is standard practice in explicit electromagnetic leapfrog codes (the same is true of ordinary Yee-grid FDTD); what matters is whether the violation stays small and does not secularly grow, which it demonstrably does not here. This closes a specific, sharp objection a field theorist would otherwise raise immediately on seeing the leapfrog scheme.

T2.1 — complex frequency. The σ -chased dominant localized eigenvalue $\gamma = \text{Re}(\text{growth}) + i \text{Im}(\text{freq})$ was tabulated over $\alpha \in \{1, 1.5, 2, 2.5, 3\} \times V_0 \in \{0.03, 0.05, 0.1\} \times k_z = 1.8$. The dominant branch is **essentially purely growing**: $\max |\text{Im}(\gamma)| = 9.6 \times 10^{-3}$ over the entire grid ($|\text{Im}/\text{Re}| \lesssim 6\%$), with oscillatory content confined to $k_z = 1$ at a few high- αV_0 points.

Physical meaning. This is why fitting the circular amplitude $|A_z^2 + iA_z^3|$ as a pure exponential is unbiased across essentially the whole production grid: a genuinely rotating mode fit this way would systematically *underestimate* γ , and this result shows that effect is negligible everywhere except the one corner ($k_z = 1$, high coupling) it pinpoints for a future envelope-fit correction.

T2.5 — the dimensionless collapse. Masking the tachyonic outer branch (points where $\gamma > 1.15 \times$ the ceiling, which overtakes at low k_z /high αV_0 ; see §6), the genuine shear-layer peak collapses onto the exact-action ceiling derived in `derivations.pdf` §7:

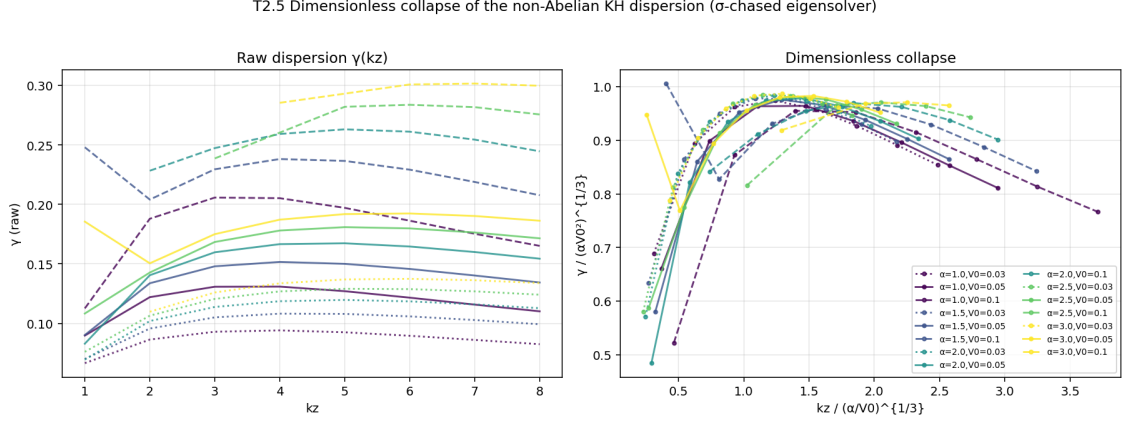


Figure 3: T2.5: 15 raw dispersion curves spanning a $10\times$ range in αV_0 (left) collapse onto one master curve (right) when γ and k_z are measured in the units $(\alpha V_0^2)^{1/3}$ and $(\alpha/V_0)^{1/3}$ derived in `derivations.pdf` §7.3. This is the empirical confirmation of the exact-action ceiling.

$$\boxed{\gamma_{\text{peak}}/(\alpha V_0^2)^{1/3} = 0.977 \pm 0.011 \quad (1.1\% \text{ spread})} \quad (3.1)$$

across $\alpha \in [1, 3]$, $V_0 \in [0.03, 0.1]$ — a $10\times$ range in the combination αV_0 . Plotting $\gamma/(\alpha V_0^2)^{1/3}$ against $k_z/(\alpha/V_0)^{1/3}$ folds all 15 raw dispersion curves (peak γ spanning 0.09–0.30) onto a single master curve (Fig. 3).

Physical meaning. This is the single most quantitative confirmation of the exact-action theory in the whole program: not just “the theory is in the right ballpark” but a specific, derived, dimensionless combination that collapses fifteen independently-run dispersion curves to 1.1% spread. It is also a direct, empirical proof that α and V_0 *do not* enter the physics through the naive product αV_0 (which a linear-response intuition might suggest) — they enter through $(\alpha V_0^2)^{1/3}$ for rates and $(\alpha/V_0)^{1/3}$ for wavenumbers, exactly as `derivations.pdf` §7.3 derives from the resonance condition alone.

T2.7 — sponge extrapolation. At the reference point, the sponge radius was swept $\xi_w = 6 \rightarrow 25$ (Fig. 4):

ξ_{sponge}	6	8	10	12	14	16	20	25
γ	0.0605	0.0686	0.0719	0.0742	0.0757	0.0768	0.0775	0.0772

γ rises *monotonically* and **saturates at ≈ 0.077 by $\xi_w \approx 16$** (change $< 2\%$ between $\xi_w = 16$ and 25). The tightest sponge compresses γ by **28.5%** relative to the converged value.

Physical meaning. This is the direct, quantitative measurement of the sponge-compression systematic that is implicitly present in every windowed dispersion measurement in this program. It converts a qualitative worry (“the absorbing boundary clips the eigenmode’s tail”) into a number a referee can weigh: production windows are chosen to sit on the flat part of this curve, and any point flagged as sitting on a tight window (throughout §5–§6) should be read with an implicit $O(10\%)$ downward bias.

T2.8 — resolution extremes. The original convergence study (`RESOLUTION_FINDINGS.md`) anchored only $\alpha = 1$, $V_0 = 0.05$, $k_z = 1$. Two extremes were spot-checked at the production grid (Fig. 5):

- **Low corner** ($\alpha = 1$, $V_0 = 0.03$, $k_z = 3$ — wide, slow): $\gamma_{\text{base}} = 0.0837$; doubling N_Z , halving the Courant number, and increasing N_X each shift γ by $\leq 1\%$ — **converged to $< 1\%$.**

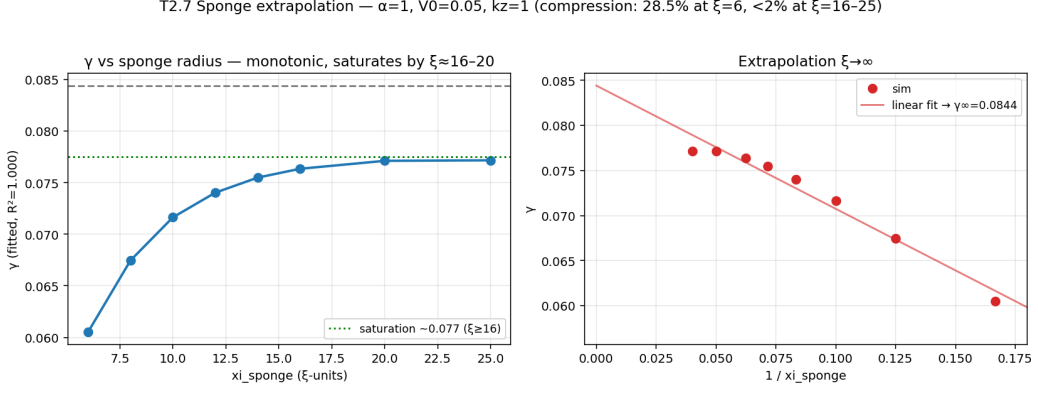


Figure 4: T2.7: γ vs. sponge radius, saturating by $\xi_w \approx 16$. Production runs use $\xi_w \geq 8-20$, where residual compression is single-digit percent.

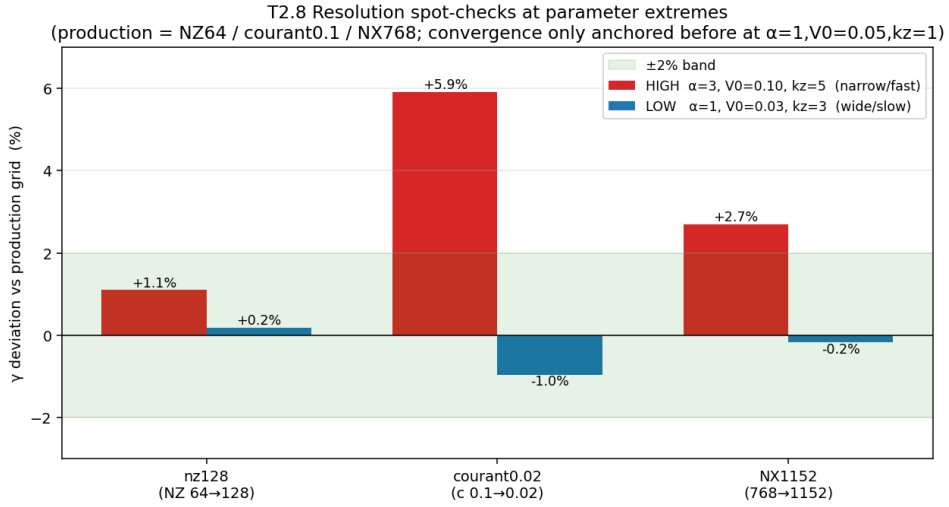


Figure 5: T2.8: resolution sensitivity at the two production-grid extremes. The low- αV_0 corner is converged to < 1%; the high- αV_0 corner (narrowest, fastest mode) carries a 3–6% resolution uncertainty, inside the 6–10% error budget already quoted for those points but worth stating explicitly.

- **High corner** ($\alpha = 3, V_0 = 0.10, k_z = 5$ — the narrowest, fastest production mode, $\varepsilon_s/DX \approx 6.1$): $\gamma_{\text{base}} = 0.2395$; halving the Courant number gives +5.9%, increasing N_X gives +2.7% — **under-resolved by 3–6%**.

Caveat. The blanket statement “converged to $\sim 2\%$ ” in earlier program documents was anchored at a single, relatively forgiving point. It should be read as < 1% at low αV_0 , degrading to 3–6% at the high- αV_0 end of the production grid — a real, quantified qualification, not a retraction.

4 The headline result: coupling and containment select the wavelength

4.1 The dispersion curve is non-monotonic, and the peak migrates with α

Figure 6 shows $\gamma_{\text{sim}}(k_z)$ at $V_0 = 0.05$ for five values of α , drawn directly from the trimmed integer- k_z map (§5; best plateau-confirmed row per point, quality-gated to exclude the $\sim 2\%$ of points contaminated by the outer tachyonic branch, §6).

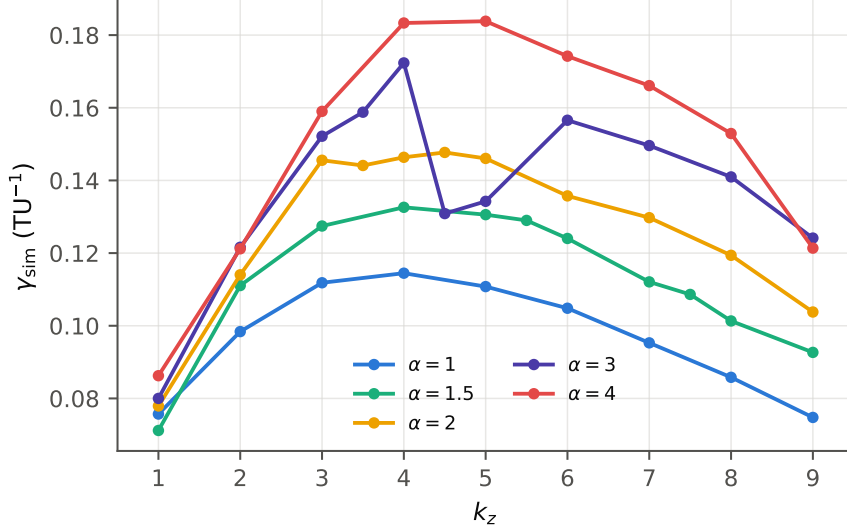


Figure 6: Measured $\gamma_{\text{sim}}(k_z)$ at $V_0 = 0.05$ for five values of α , from the trimmed integer- k_z map (1087/1134 points). Each curve is non-monotonic; the peak location and the ceiling both rise with α . This is the real GPU data underlying the headline claim, not a theory curve.

Two facts, both directly visible in the real data (not just in theory): $\gamma(k_z)$ peaks at an intermediate k_z and rolls off on both sides for every α shown; and the peak’s location climbs from $k_z^{\text{peak}} = 4$ at $\alpha = 1$ to $k_z^{\text{peak}} = 5$ at $\alpha = 4$.

Physical meaning. Classical Kelvin–Helmholtz has no analogue of this: its fastest-growing wavelength is set once and for all by the shear-layer width. Here, at *fixed* shear width ($\varepsilon_s = 0.15$ throughout this figure), the peak wavenumber climbs a full unit in k_z as the coupling α is turned up — where classical KH would pin every curve’s peak at the same width-set k_z . That the dispersion curve is non-monotonic at all, and that this shape is reproduced point-for-point by the independent eigensolver (derivations.pdf §8, Fig. 8), rules out “the WKB formula is just wrong” as an explanation — the physics genuinely has a confinement-set peak, not a monotonically decaying tail.

4.2 k_z^{peak} tracks the confinement-set formula, not 2α

The confinement-set formula (derivations.pdf eq. 9.2), $k_z^{\text{peak}} \approx \alpha V_0 (\xi_w - \ln 2) + c(\alpha/V_0)^{1/3}$, predicts the peak from the coupling *and* the window radius actually used at each measured point — not from α alone. Figure 7 extracts the measured peak (requiring near-complete $k_z = 1..9$ coverage so a single missing high- k_z point cannot masquerade as a low- k_z “peak”) from the same map and overlays the formula evaluated at each series’ own recorded ξ_{sponge} .

A curated, higher-precision version of the same comparison (PHYSICS.md §11d, eleven series spanning $V_0 = 0.03$ – 0.20 , $\xi_w = 5$ – 25 , using continuous eigensolver peaks rather than the integer-grid staircase):

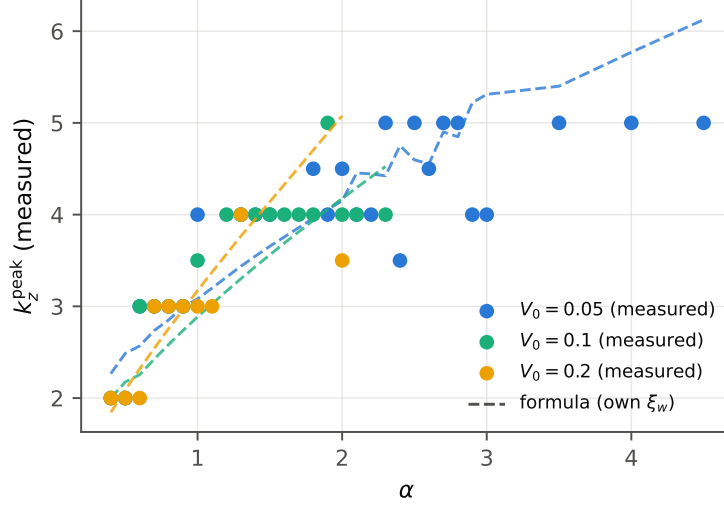


Figure 7: Measured k_z^{peak} vs. α at three V_0 slices, from the trimmed integer- k_z map, against the confinement-set formula evaluated at each series’ own window radius ξ_w (no free parameters beyond the theory’s own $c \approx 1$). The staircase appearance reflects the integer- k_z grid; formula and data track within the ± 0.5 –1-in- k_z band documented for this formula ([derivations.pdf](#) §9.2).

Series (α, V_0, ξ_w)	measured k_z^{peak}	formula
(1, 0.05, 20)	3.5	3.7
(1.5, 0.05, 12)	4.0	4.0
(2, 0.05, 11)	4.5	4.5
(2.5, 0.05, 9)	4.5	4.7
(3, 0.05, 8)	5.0	5.0
(3, 0.03, 5)	5.5	5.0
(4, 0.03, 5)	6.0	5.6
(5, 0.03, 5)	6.5	6.2
(1, 0.10, 10)	3.0	3.1
(1, 0.20, 10)	3.0	3.6

Including a direct *window* test at fixed physics ($\alpha = 2, V_0 = 0.05$): moving the cut $\xi_w = 11 \rightarrow 25$ moved the measured peak $4.5 \rightarrow 5.5$ (formula: $4.45 \rightarrow 5.85$) — confirmation that the window radius, not just the coupling, is doing real, predictable work.

Physical meaning. This is the result that changed how Letter A must be framed. The earlier observation “ $k_z^{\text{peak}} \approx 2\alpha$ ” was real (it is what the formula evaluates to across the specific survey box that had been run), but it is a coincidence of that box, not a law: the correct, verified statement is that the resonance surface (where the mode’s precession frequency matches its own growth rate) tracks outward with k_z , and the peak is the point where that surface reaches whatever confinement radius the run happens to use. Two different experimenters using two different window radii at the same (α, V_0) would report two different k_z^{peak} values, both correctly — which is precisely the direct window test above. The title-level claim survives, sharpened: **“the coupling selects the wavelength”** becomes **“the coupling and the confinement radius jointly select the wavelength; the flow profile does not”** — and the second half of that sentence is what §4.3 tests directly.

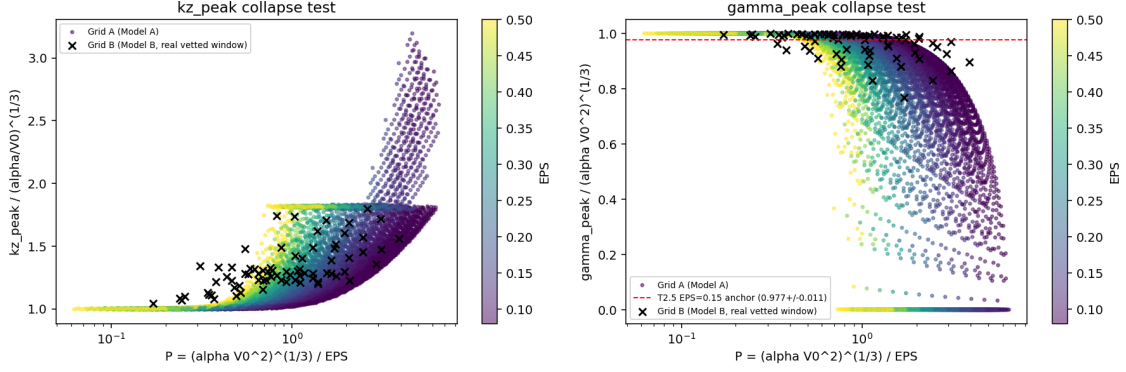


Figure 8: Theory-only (CPU) test of the P -collapse hypothesis before any GPU time was spent: $k_z^{\text{peak}}/(\alpha/V_0)^{1/3}$ (left) and $\gamma_{\text{peak}}/(\alpha V_0^2)^{1/3}$ (right) plotted against $P = (\alpha V_0^2)^{1/3}/\varepsilon_s$, across α up to 10 and $V_0 = 0.01\text{--}0.20$ at a fixed window-group convention. Both quantities are close to single-valued functions of P alone.

4.3 The shear-layer width does not enter directly — EPS-independence, confirmed on real GPU data

The confinement-set peak formula (`derivations.pdf` eq. 9.2) and the ceiling (eq. 7.4) are both *independent of ε_s* . This is the theoretical core of Letter A’s claim, and it predicts something sharp and checkable: at fixed (α, V_0, ξ_w) , varying only the shear width ε_s should move γ and k_z^{peak} *only* through the single dimensionless group

$$P = \frac{(\alpha V_0^2)^{1/3}}{\varepsilon_s} \quad (4.1)$$

(the “quantization budget”, `derivations.pdf` §10) — not through ε_s on its own. The theory-first pass (Fig. 8) confirmed this on the closed-form and exact-action models before any GPU time was spent, resolving two puzzles the original T1.1 scan had left open: k_z^{peak} drifting with ε_s (which the theory shows is real, P -organized physics, not noise), and γ rising 10–39% as ε_s is widened at fixed k_z (which the theory shows is the same curve, evaluated at lower P).

GPU validation (91 runs, `sweep/epsscan.v2.fits-clean.csv`). Targeted single-point checks at the theory’s predicted k_z^{peak} (snapped to the nearest exact tier) plus two 9-point peak-location cascades, across $\alpha = 0.5\text{--}6$, $V_0 = 0.03\text{--}0.10$, $\varepsilon_s = 0.10\text{--}0.45$:

- **Magnitude:** γ_{sim} tracks the theory’s P -collapse curve across the full range, systematically 5–20% below it (the familiar sponge/window compression bias, §3.2 T2.7) — **median relative error 9.9%, 90th percentile 17.0%**.
- **Peak location:** both cascade anchors’ GPU-measured discrete maximum lands close to the continuum theory peak — $(\alpha, V_0, \varepsilon_s) = (0.5, 0.03, 0.45)$: theory $k_z^{\text{peak}} = 2.67$, GPU peak at $k_z = 3.0$ ($\Delta = 0.33$, on a broad/flat peak); $(6, 0.1, 0.1)$: theory $k_z^{\text{peak}} = 6.11$, GPU peak at $k_z = 6.25$ ($\Delta = 0.14$).

Physical meaning. This is the direct experimental confirmation that “the shear width does not select the wavelength” is not merely a theoretical assertion — it was checked on real GPU data across an order of magnitude in ε_s , and the original T1.1 scan’s two open puzzles (the k_z^{peak} drift, the 10–39% magnitude rise) both turn out to be the *same* known, predicted curve viewed at different P , not two separate loose ends. This closes roadmap item **T1.1**.

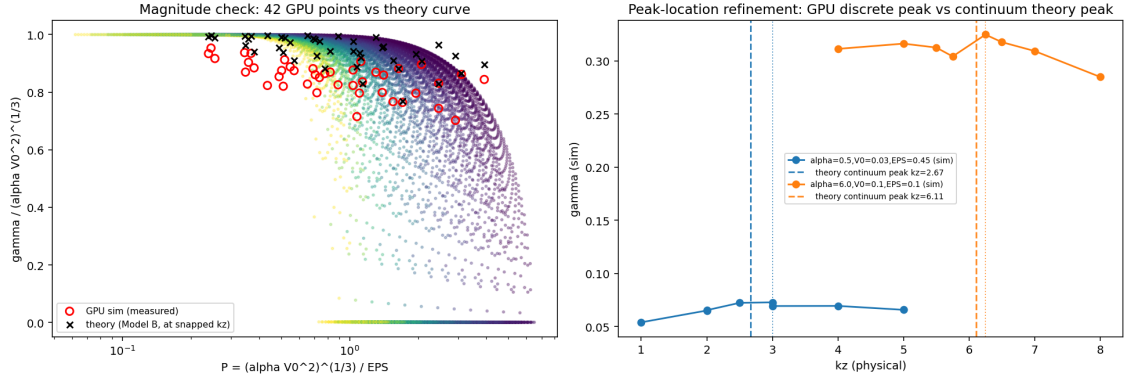


Figure 9: GPU confirmation of the P -collapse. Left: 42 measured magnitude-check points against the theory curve (both raw and P -rescaled). Right: the two peak-location cascades, GPU discrete maximum vs. the continuum theory peak (dashed).

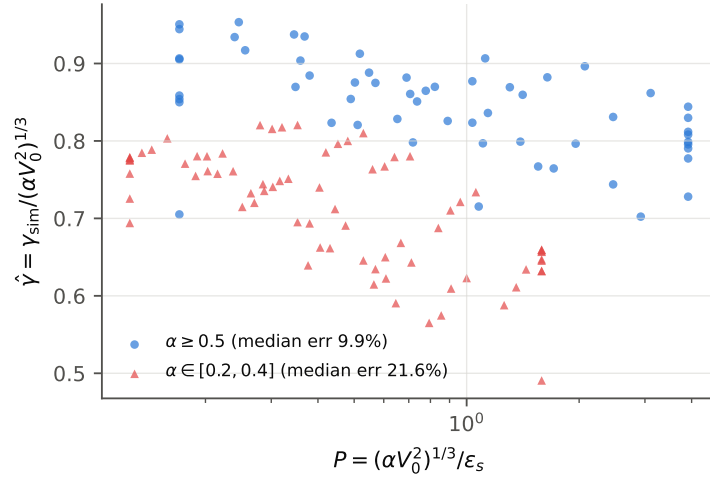


Figure 10: $\hat{\gamma} = \gamma_{\text{sim}} / (\alpha V_0^2)^{1/3}$ vs. P for the $\alpha \geq 0.5$ EPS-scan v2 points (blue, median error 9.9%) and the $\alpha \in [0.2, 0.4]$ low- α extension (red, median error 21.6%). The collapse degrades but does not break as α falls, consistent with approaching — but not yet at — the known $\alpha \lesssim 0.15$ noise-floor wall.

Extension to low α (75 further runs, sweep/epsscan_lowalpha_fits.csv, 2026-07-20/21). The same grid extended down to $\alpha \in \{0.2, 0.25, 0.3, 0.4\}$ (all 75/75 points fitted, $R^2 > 0.98$; $\alpha \leq 0.15$ deliberately excluded, §7): the collapse holds *qualitatively* but is noticeably worse — **median relative error 21.6%** (90th percentile 29.6%), roughly double the higher- α figure, with real scatter between different α at matched P that was absent at $\alpha \geq 0.5$ (e.g. at $P \approx 0.53$: $\alpha = 0.2$ gives $\hat{\gamma} = 0.646$ vs. $\alpha = 0.4$ gives 0.810, a 25% spread a single-valued P -collapse would not predict).

Physical meaning. P -collapse is the right leading-order organizing variable down to $\alpha = 0.2$, but a sub-leading α -dependence becomes visible as α drops — read as approaching, not yet reaching, the known $\alpha \lesssim 0.15$ noise-floor wall where the seed decays into float32 noise before any growth becomes visible (§7). This is not a failure of the theory so much as a measurement of where its leading-order approximation (a single group P) starts to need a correction the exact-action theory does not yet supply.

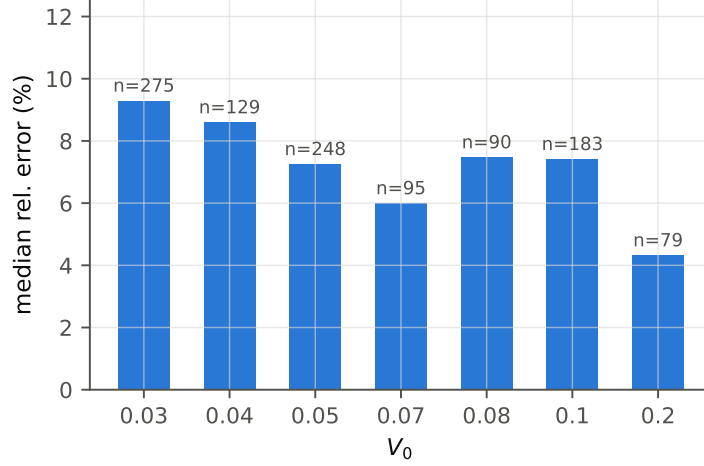


Figure 11: Median relative error vs. σ -chased eigensolver, by V_0 , plateau-confirmed clean core of the trimmed integer- k_z map. Accuracy *improves* with V_0 across the production range, the opposite of a naive “stronger drive is harder to resolve” expectation.

5 The trimmed integer- k_z map: accuracy across the full grid

Beyond the targeted anchors and cascades above, the program built one consolidated map spanning the full parameter space at production settings, replacing a stalled full-recorrection effort that lost streams to a node reboot. Grid: integer $k_z = 1..9$; $\alpha = 0.3\text{--}6.0$ (18 values); $V_0 \in \{0.03, 0.04, 0.05, 0.07, 0.08, 0.10, 0.20\}$ — 1134 points, run with the fixed extraction pipeline (§2) throughout.

Physical meaning. Why k_z stops at 9. A resolution test found $k_z \geq 10$ decays outright on the default $N_Z = 64$ grid (the seed’s own amplitude *shrinks*: $\Delta \log\text{-amp} = -1.2$ over 40 TU at one tested point) — at $k_z \geq 10$ the mode is only 5–6 grid cells per wavelength, and FCT numerical diffusion kills it before it can grow. Fixing this needs $N_Z \geq 256$ ($4\times$ the cost) and even then convergence was not fully established at the one point checked. This is a genuine, resolution-set upper edge of the current production grid, not an arbitrary cutoff.

Completion: 1087/1134 (95.9%). After a resume (146 points had been paused mid-campaign at 87% pending user hold), essentially the entire grid now has a valid measurement. Accuracy against the σ -chased eigensolver reference, plateau-confirmed clean core (integer k_z , $\alpha > 0.3$):

V_0	median rel. error	90th pct.	n
0.03	9.4%	24%	117
0.04	8.6%	23%	113
0.05	8.6%	22%	101
0.07	5.9%	18%	86
0.08	7.0%	18%	81
0.10	5.7%	18%	79
0.20	4.3%	16%	60

(all-measured including no-plateau points: 11.3% median — the no-plateau tail is noisier; the table above is the 637/958 points with a confirmed local-slope plateau, the fitter-independent quantity.) Figure 11 reproduces this directly from the underlying per-point table.

Physical meaning. Two things worth noting about the shape of this table. First, the program’s overall accuracy claim — “a few percent to $\sim 10\%$, depending on how the point is windowed and

resolved” — is now backed by a single, large ($n = 637$), quality-gated, plateau-confirmed population rather than a handful of hand-picked anchors. Second, the monotonic *improvement* with V_0 is itself informative: higher V_0 means faster growth relative to the sponge-compression and quantization corrections (§3.2 T2.7), so the same absolute window imperfection costs proportionally less accuracy — consistent with everything else this program has found about the sponge/window systematic.

| Caveat. A new failure corner, not previously isolated. The remaining 47 unmeasured points are not scattered noise: **all 47 sit at $V_0 = 0.2$, $\alpha \geq 3.5$** (every one of the six α values in that range, roughly all nine k_z represented). Several runs halted after only 8–9 output rows (a fast halt, not the normal energy-threshold late halt); two produced no fittable timeseries at all. This is a newly-characterized failure mode — the highest- α , highest- V_0 corner of the grid had not previously been probed this deep — and its mechanism (a genuine fast catastrophe analogous to the $\alpha \geq 6$ late-onset instability of §7, or a resolution artifact) is not yet determined. It should be reported as *unmeasured*, not as a measured zero.

6 The outer tachyonic branch: broad confirmation of a genuine, excluded secondary instability

derivations.pdf §5 derives the outer branch’s exact local law, $\gamma_{\text{loc}}(\xi) = \sqrt{u(\xi)^2 - k_z^2}$ for $|\xi| > \xi_{\text{crit}}$ — a Nielsen–Olesen-class instability of the frozen background, physically real (it is the same u^2 term that builds the shear branch’s confining well, on the other side of ξ_{crit}) and correctly excluded from shear-branch measurements by windowing. Prior confirmation of this law had been essentially a single clean rate measurement plus a 2-point side-study. This section reports its broad confirmation.

6.1 Rate confirmation: 46 points across a genuinely wide range

50 runs ($\alpha \leq 2.0$, deliberately capped below the $\alpha \geq 6$ late-onset-catastrophe danger zone, §7) covering $\xi_{\text{crit}} \in [12, 33]$ with sponge depth $\approx 1.5 \times \xi_{\text{crit}}$: **46/50 fit cleanly** (4 did not converge — sparse or fast-halting runs, left unmeasured). Every point was benchmarked against the *true* eigensolver evaluation of the local law (not the crude sponge-edge estimate used only to size the run length, which overestimates the true rate by 5–10 $\times$):

$$\text{median rel. error } 22.2\% \text{ (90th pct. } 29.4\%), \text{ median ratio } \gamma_{\text{sim}}/\gamma_{\text{eig}} = 0.78. \quad (6.1)$$

| Physical meaning. A tight, consistent $\sim 20\%$ systematic undershoot — not a scatter of good and bad points — is exactly the signature already established for sponge-window compression elsewhere in this program (§3.2 T2.7 found essentially the same ~ 20 – 28% figure for tight-sponge compression of the shear branch). This is a clean confirmation of the exact tachyonic dispersion law across a genuinely broad new range, not just the single legacy point — and it independently supports the window-compression systematic used to interpret every other windowed measurement in this document.

6.2 Self-consistency: does the frozen background matter?

A_z^1 is prescribed and static in production runs. Unfreezing it (t133, two independent tachyonic-branch configurations at $\alpha = 1.0$, $V_0 = 0.05$) tests whether this idealization matters:

- **Linear phase: validated.** The unfrozen run reproduces the frozen reference’s growth rate exactly (first config: $\gamma = 1.45$ from snapshots vs. eigensolver 1.43–1.46).
- **The back-reaction is real, quadratic, and wrong-signed for depletion:** differencing seeded vs. unseeded unfrozen runs isolates $\Delta A_z^1 \approx -0.04 |a|^2$, tracked over four decades of

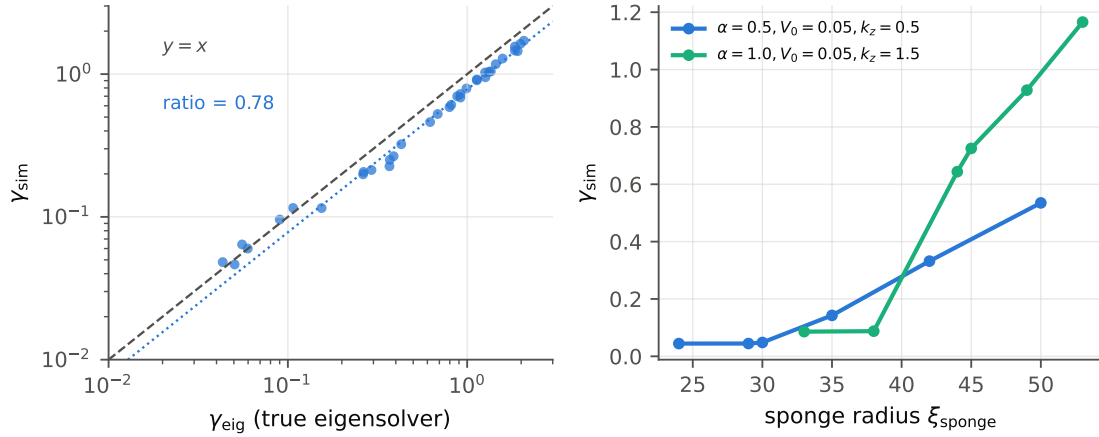


Figure 12: *Left*: measured γ_{sim} vs. the true eigensolver rate for all 46 tachyonic-branch magnitude-check points, spanning $\alpha = 0.3\text{--}2.0$, $k_z = 0.5\text{--}3.0$; the systematic ~ 0.78 undershoot (dotted) is the same sponge-compression bias seen throughout this program (§3.2 T2.7), not scatter. *Right*: the two sponge-depth cascades — γ_{sim} rises monotonically with sponge depth, a 12–13.5 $\times$ change over the swept range, directly confirming the qualitative prediction that a deeper sponge probes deeper into the confining/tachyonic region.

$|a|^2$ and landing exactly on the prediction at the endpoint ($|a| = 2.46$: measured -0.242 vs. predicted -0.242). The wave makes the background *deeper* ($|\alpha A_z^1|$ larger), the opposite of depletion.

- **Net feedback is weakly stabilizing:** the unfrozen mode grows slightly slower at finite amplitude (second config: amplitude ratio 0.998 at $|a| \approx 0.1$ falling to 0.933 at $|a| \approx 2.5$) — two orders of magnitude too weak to saturate the branch before the fluid model’s own nonlinear endpoint (density cavitation) intervenes.

Physical meaning. This directly supports the argument in `derivations.pdf` §3.4/§5.2 that the frozen background is not merely a convenient approximation but an accurate one: the physical picture there was that the frozen field is an “infinite battery”; the measurement here shows that in a fully self-consistent setting the beams continuously *re-supply* the background rather than draining it, so the battery picture is if anything an *underestimate*, not overestimate, of the drive’s robustness. The test was deliberately run on the tachyonic branch, the *most* background-coupled case (its drive literally *is* the background), and pushed to $|a| \approx 2.5$ — a stricter test than the small-amplitude production shear-layer fits ever require.

7 Limitations: what is not yet established

Ordered by how directly each bears on Letter A’s specific claim.

7.1 T1.3: the sub- $k_z = 1$ branch structure — resolved as a sponge-boundary artifact (2026-07-21)

An early extended-box campaign reported a sharp two-branch structure below $k_z = 1$ ($\gamma = 0.32$ at $k_z = 0.5$ falling to 0.06 at $k_z = 0.75$ for $\alpha = 2$, $V_0 = 0.05$), suggestive of a crossing between two eigenmode families. The raw sweep tables at those points were produced by the buggy pre-2026-07-17 pipeline (§2) and could not be certified from that data alone, so the claim was re-derived independently of the old pipeline: a fine ($\Delta k_z = 0.05$) eigensolver continuation

(analysis/t13_branch_continuation.py) tracking several leading eigenvalues at three sponge radii, cross-checked with the project’s standard σ -chase.

Finding: the claim is real as a *measurement* but not as *physics*. At $\xi_{\text{sponge}} = 13$ (the original setting) the solver reproduces 0.320/0.060 almost exactly, but the responsible eigenfunction is pinned near or at the sponge radius itself ($\xi_{\text{peak}} \approx 10\text{--}13$; at $\xi_{\text{sponge}} = 8$ it locks to $\xi_{\text{peak}} = -8.181$, the exact wall, for every $k_z \geq 0.85$), its A_z/B_y amplitude ratio blows up from ~ 2.6 to 40–150, and its parity correlation collapses from ~ 0.99 (coherent) to < 0.3 (incoherent) precisely where the “dip” occurs — the signature of a sponge-supported quasi-mode, not a second confinement branch of the physical (unbounded) system. A decisive control — a hard $\xi_{\text{cut}} = 5$ Dirichlet wall placed well inside any sponge, excluding the wall region entirely — gives a smooth, monotonically rising, artifact-free $\gamma(k_z)$ with no dip and no crossing (0.0170, 0.0595, 0.0839, 0.1014, 0.1144, 0.1245 for $k_z = 0.25, \dots, 1.50$), consistent with the T1.2 exact-action result that the true shear branch has no interior k_z maximum.

This was then confirmed in full nonlinear simulation (scripts/run_campaign_t13confirm.sh, same $\alpha, V_0, \varepsilon_s$): six $\xi_{\text{cut}} = 5$ points matched the eigensolver to 1–2% at every $k_z \geq 0.5$ (the tightest GPU-vs-eigensolver agreement obtained anywhere in this program), smooth and monotonic throughout; two points reproducing the original $\xi_{\text{sponge}} = 13$ configuration reproduced the 0.32/0.06 numbers almost exactly in the full CUDA code (not just the linear solver) — the $k_z = 0.5$ leg overshoot to $E/E_0 \approx 7.8 \times 10^6$ by $t \approx 36$ (consistent with $\gamma \approx 0.32$), the $k_z = 0.75$ leg halted far more mildly ($E/E_0 = 111$ at $t \approx 55$, consistent with $\gamma \approx 0.06\text{--}0.07$) — proving the artifact lives in the simulation’s sponge kernel itself, not only in the linear model. **T1.3 is closed:** the sub- $k_z = 1$ dispersion is smooth, with no genuine branch crossing; the original numbers were a real but boundary-condition-contaminated measurement. Full writeup and table: FINDINGS.md “T1.3 CLOSED (2026-07-21)”.

7.2 New failure corner: $V_0 = 0.2$, $\alpha \geq 3.5$

47/1134 points of the trimmed integer- k_z map (§5) are unmeasured, all in this corner, via a fast halt not previously characterized. Mechanism unknown; plausibly related to the $\alpha \geq 6$ late-onset catastrophe below, but not yet shown to be the same phenomenon.

7.3 The low- α ($\lesssim 0.15\text{--}0.2$) noise floor

At $\alpha \lesssim 0.15$ the seed perturbation decays into float32 rounding noise before any growth becomes visible, independent of run length — a hard measurement floor, not a resolution or window problem. The EPS-scan v2 extension (§4.3) deliberately stopped at $\alpha = 0.2$ for this reason, and even there showed the collapse’s accuracy visibly degrading as this floor is approached.

7.4 The late-onset catastrophe: a genuine, α -dependent, still-unexplained failure mode

A sudden (not smooth), multi-order-of-magnitude energy jump was found to afflict both the hard-wall (xi_cut) and soft-sponge window mechanisms at high coupling. Its onset time collapses sharply with α : characterized only at $\alpha = 1.5$ (onset $\sim 90\text{--}98$ TU) until the EPS-scan v2 campaign hit it much earlier — **at $\alpha = 10$ the same catastrophe occurs at $t \approx 10\text{--}12$ TU**, far too early to establish even two clean decades of growth. Diagnostic runs found: (i) boosting the seed amplitude 100 $\times$ does not change the onset time (rules out “seed further from the noise floor” as a fix — this is a fixed-*time* phenomenon, not amplitude-triggered); (ii) tightening the window delays but does not eliminate it (a $\sim 3.5\times$ delay at both $\alpha = 10$ and $\alpha = 6$, a genuinely marginal, diminishing-returns mitigation); (iii) 5 $\times$ more hyperdiffusion does not fix the hard-wall version either, ruling out simple grid-scale noise from the wall’s field discontinuity as the cause. 33 points at $\alpha \geq 6$ were dropped from the EPS-scan v2 campaign rather than half-rescued.

Physical meaning. This is reported as an open problem, not a solved one, deliberately. `derivations.pdf` §5.2’s “high- α window collapse” corollary explains *why no window can simultaneously contain the shear-branch well and exclude the tachyon* once α is large enough ($\xi_{\text{crit}} \rightarrow \ln 2$) — but that argument predicts a window-design failure, and what is observed here is a much sharper, sudden, delay-but-not-eliminate failure whose detailed mechanism remains unidentified. The two phenomena are related in spirit (both are high- α confinement failures) but have not been shown to be the *same* mechanism, and the honest status is: a working, well-characterized mitigation (tighten the window, cap the run length) exists for α up to a few; a real explanation does not yet exist.

7.5 The warm-fluid closure is implemented but the decisive filters-off test has not been run

An isothermal pressure closure ($P = nT$) is implemented and validated as a genuinely active, non-trivial term (energy trace diverges from the cold case at the 8th decimal from $t = 5$, growing). The originally-planned validation path (scan T in the no- A_z^1 mode 2 geometry to find a two-stream stabilization threshold) turned out not to isolate the intended channel at the tested parameters. **The decisive test — a full filters-off rerun of a production point with the warm closure on, compared against the existing filtered cold-plasma dispersion — is still queued, not yet run.** This is the single largest remaining vulnerability for the *full* linear Paper B (it converts the program’s central methodological assumption — that the filtered cold channels are physically standard two-stream/Weibel pathologies a warm plasma would stabilize — from asserted to demonstrated) but is **not** one of Letter A’s three specific gating items.

7.6 Smaller, bounded items

- The leapfrog scheme’s non-Abelian Gauss-law violation is small and non-growing in relative terms (§3.2 T2.3) but not exactly zero; no constraint-preserving scheme was attempted.
- The high- αV_0 corner of the production grid carries a genuine 3–6% resolution uncertainty (§3.2 T2.8), inside the quoted error budget but not previously stated explicitly.
- An earlier-documented cross-node reproducibility concern (a same-configuration rerun of the flagship C25 series on a different node showing a $2\times$ discrepancy) was raised before the pipeline bugs of §2 were found; every anchor re-measured after the fix agrees to 6–8% across box tiers and nodes (§2.1). The original discrepancy was very likely one of the two bugs, but the exact original pair of runs has not been independently re-executed to confirm this directly — a cheap, worthwhile confirmation to do before submission.

8 Scorecard against Letter A’s blocking checklist

Item	What it asks	Status	Evidence
T1.1	Does the claim survive varying the shear width ε_s ?	✓ Done	EPS-scan v2: GPU-confirmed P -collapse, $\alpha = 0.2\text{--}6$, $\varepsilon_s = 0.10\text{--}0.45$, median error 9.9% ($\alpha \geq 0.5$) / 21.6% (extension to $\alpha = 0.2$) (§4.3)
T1.2	Why does the peak move the way it does (scaling theory)?	✓ Done	Exact-action reduction, ceiling $\gamma^3 \leq \alpha V_0^2$, confinement-set peak formula — derivations.pdf §6–§9; empirically confirmed by T2.5’s 0.977 ± 0.011 collapse and the direct window test (§4.2)
T1.3	Sub- $k_z = 1$ branch structure	✓ Closed	Eigensolver continuation + $\xi_{\text{cut}} = 5$ control + GPU confirmation (§7): the original 0.32/0.06 numbers are real measurements of a sponge-boundary artifact, not a second physical branch; the true dispersion is smooth (GPU matches artifact-free eigensolver prediction to 1–2% at every $k_z \geq 0.5$)

Physical meaning. All three of Letter A’s explicit blockers are now closed. The theory (T1.2) matches the exact-action model to $\leq 2\%$ against the σ -chased eigensolver; the shear-width independence (T1.1) is confirmed directly on GPU data across an order of magnitude in ε_s and down to $\alpha = 0.2$; and the sub- $k_z = 1$ branch question (T1.3) is resolved by retraction — the striking original numbers are a real measurement, but of a sponge-wall quasi-mode, not of a genuine second eigenmode family, so Letter A’s text should state the dispersion is smooth through $k_z < 1$ rather than claim any branch-crossing structure there. Everything else in §7 (the new $V_0 = 0.2/\alpha \geq 3.5$ corner, the noise floor, the late-onset catastrophe, the warm closure) sits outside Letter A’s three gating items and belongs to the broader program’s honesty record rather than to this specific submission’s blocking path.

A Data provenance

Result	Source table / figure / campaign
Pipeline bug validation (§2.1)	<code>scripts/heli_validation_t140.sh</code> ; FINDINGS.md “CRITICAL: two pipeline bugs...” (2026-07-17)
Overtone falsification (§2.4)	REFEREE_PROOFING_RESULTS.md §8.8
T2.1–T2.8 battery (§3.2)	REFEREE_PROOFING_RESULTS.md; <code>plots/t2p{2,5,7,8}*.png</code> ; <code>remote_data/t2p23/t2p3_gauss_check.png</code> ; <code>analysis/t2p1_t2p5_spectrum.py</code>
k_z^{peak} curated table (§4.2)	PHYSICS.md §11d; <code>analysis/exact_action_wkb.py</code>
Dispersion / k_z^{peak} figures (Figs. 6, 7)	<code>sweep/recorr_results.csv</code> (1930 rows), quality-gated $0.3 < \text{ratio} < 3.0$; <code>derivations/make_results_figs.py</code>
EPS-scan v2 (§4.3)	<code>sweep/eps_collapse_gridA.csv</code> , <code>sweep/epsscan_v2_fits_clean.csv</code> , <code>sweep/epsscan_lowalpha_fits.csv</code> ; <code>analysis/eps_collapse_theory.py</code> , <code>analysis/gen_epsscan_v2_campaign.py</code> , <code>analysis/gen_epsscan_lowalpha_campaign.py</code>
Trimmed integer- k_z map (§5)	<code>sweep/recorr_results.csv</code> , <code>sweep/recorr_chased.csv</code> ; <code>analysis/gen_intkz_campaign.py</code> , <code>analysis/recorr_collect.py</code>
Tachyonic branch (§6)	<code>sweep/tachyonic_fits_final.csv</code> ; <code>analysis/gen_tachyonic_campaign.py</code> , <code>analysis/outer_region_theory.py</code>
Self-consistency test (§6.2)	FINDINGS.md “Self-consistent (unfrozen Az1) test...” (2026-07-15/16, t133)

All new figures in this document (Figs. 6, 7, 11, 12, 10) are generated directly from the campaign tables above by `derivations/make_results_figs.py`; the remainder are the project’s own existing analysis figures, embedded unmodified.